



NXP Model-Based Design Bootcamp

Lab Guide: Pre-requisite Installation

September 2026

Objectives

These are the pre-requisite steps to install and prepare your PC, before the NXP Model-Based Design Bootcamp.

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Software Requirements:

For attending the NXP Model-Based Design Bootcamp the following software is required:

- Windows 10/11 Machines
- MATLAB R2026a (Please check the MathWorks Requirements [here](#))
- [Model-Based Design Toolbox for S32K3 v1.8.0](#)
- [FreeMASTER Tool v3.2 or newer](#)
- [PEmicro Hardware Interface Drivers](#)

The following chapters guide on how to install the Software, required for attending the Bootcamp.



1. Install MATLAB & Simulink R2026a

To attend this workshop, all participants are required to install on their machine MATLAB version R2026a and Simulink, alongside additional toolboxes from MathWorks, before the session takes place.

Due to the licensing requirements for MATLAB and Simulink, MathWorks will provide trial licenses to all participants, valid before and throughout the NXP Model-Based Design Bootcamp.

Requested and provided toolboxes are:

- MATLAB
- Simulink
- Stateflow
- MATLAB Coder
- Simulink Coder
- Embedded Coder
- Simscape
- Simscape Electrical
- AUTOSAR Blockset



Install MATLAB R2026a

Note: To download MATLAB and Simulink, MathWorks requires to log in on the MathWorks website. Please create an account or login with an existing account on www.mathworks.com!

To download the trial license and install MATLAB, please follow the next steps:

1. Access the link below and press **Download Software**:

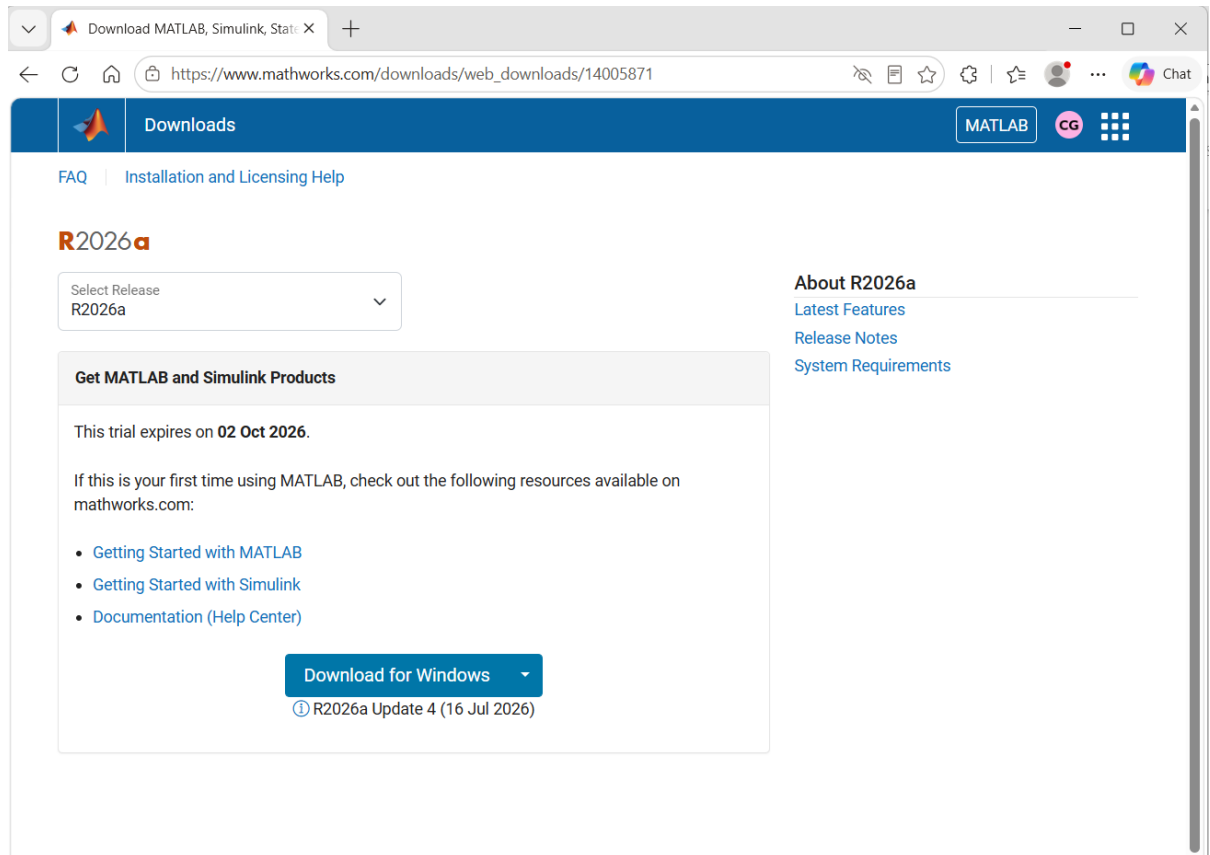
<https://www.mathworks.com/licensecenter/classroom/5477802/>

The screenshot shows a web browser window with the MathWorks License Center page. The address bar shows the URL: <https://www.mathworks.com/licensecenter/classroom/5477802/>. The page header includes the MathWorks logo, navigation links (MATLAB Help Center, Community, Learning), and a MATLAB logo. The main content area is titled "Access MATLAB for your Hands-On Workshop" and contains the following text: "MathWorks is pleased to provide a special license to you as a course participant to use for your Hands-On Workshop. This limited license is available until 02 Oct 2026 and is intended to be used only for course work and not for government, research, commercial, or other organization use." Below this text is a table with the following information:

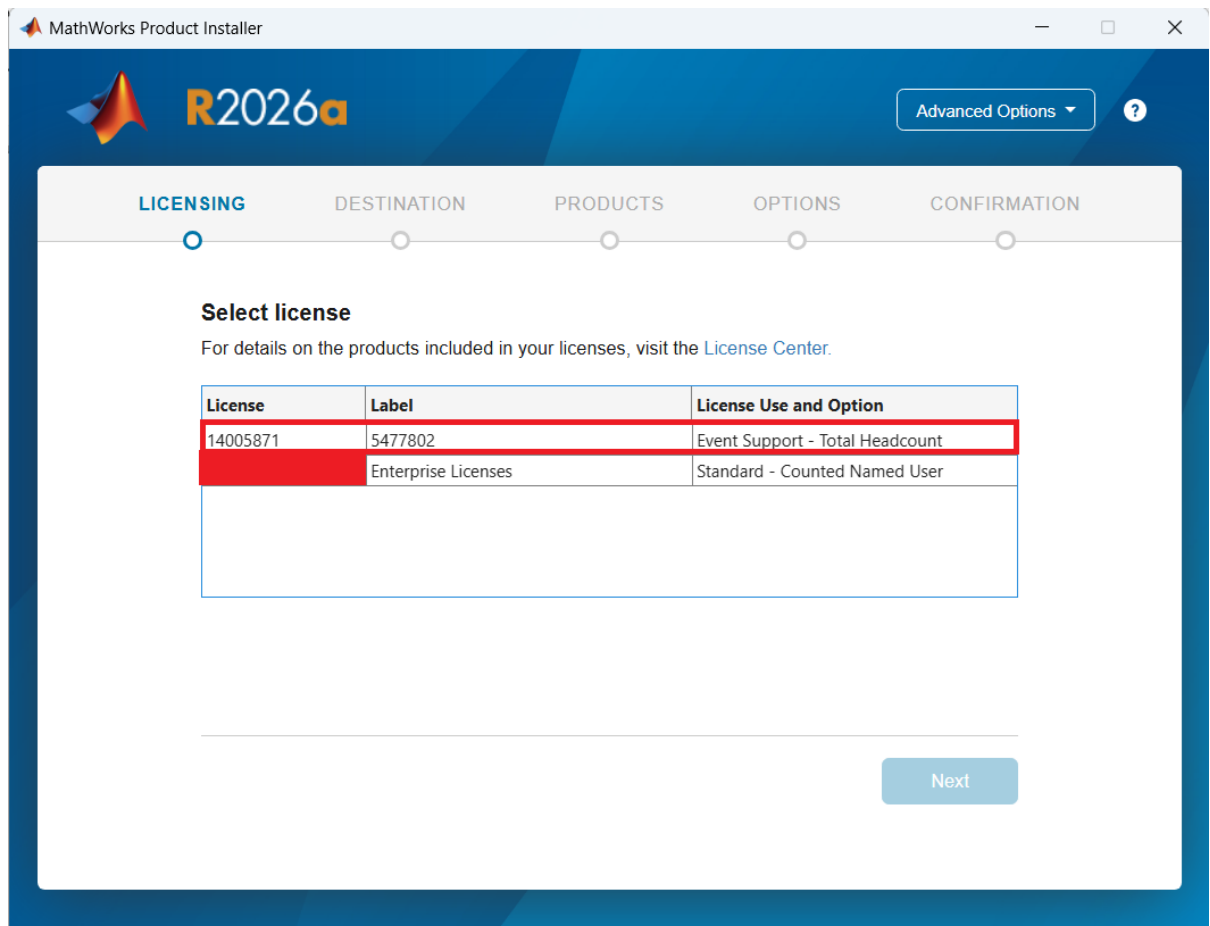
Course Name:	NXP Hands On Workshop @ University Politehnica of Bucharest
Organization:	MathWorks
Starting:	21 Sep 2026
Ending:	25 Sep 2026

At the bottom of the table is a blue button labeled "Download Software".

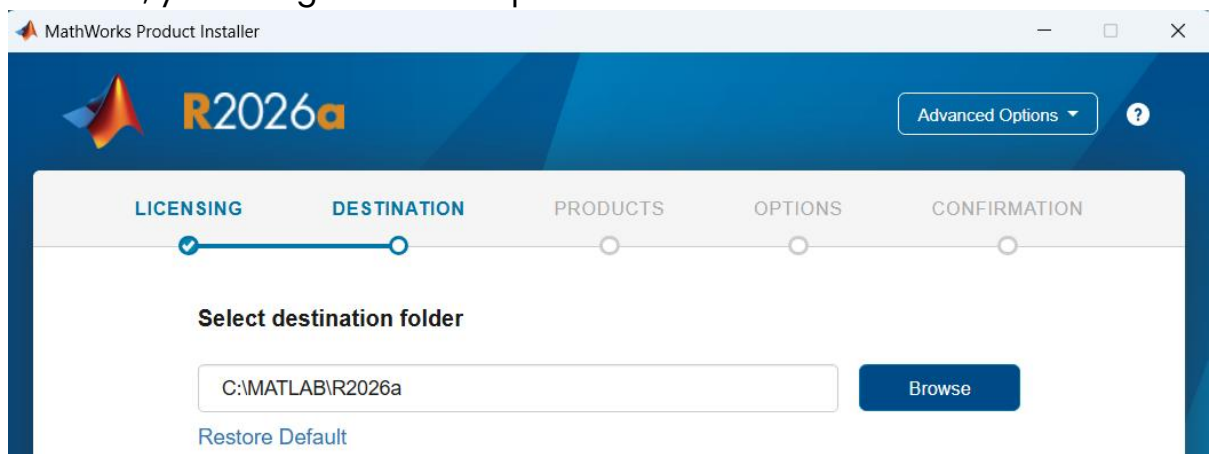
2. Make sure the **R2026a** version is selected and press the **Download for Windows** button. This action is downloading the installer in the browser's download folder.



3. Run the matlab_R2026a_Windows.exe installer, Log In and Review and accept the license.
4. Select the **Event Support** License which has been automatically linked to your account when the link from point 1 was accessed.

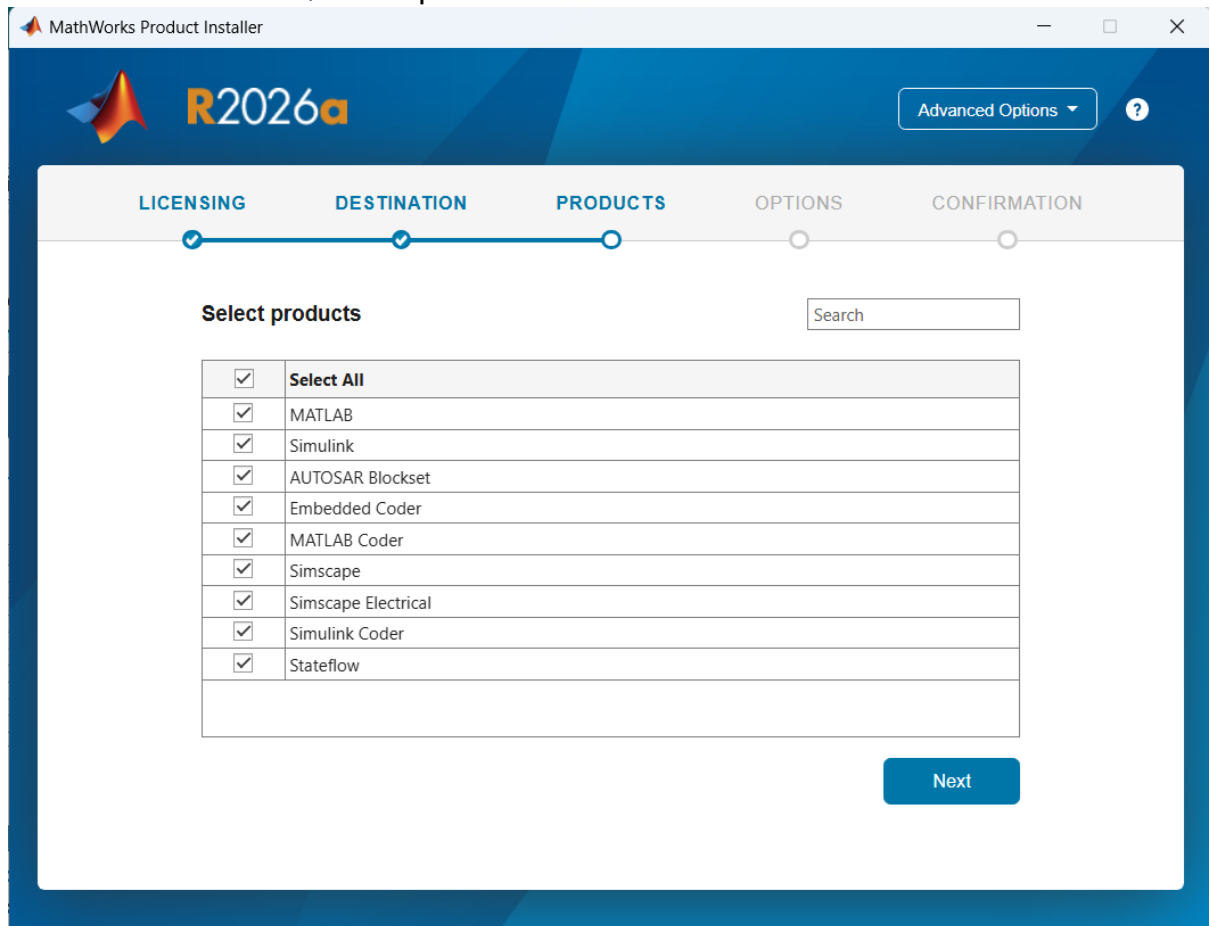


- Change the default installation path in a place with no spaces, like for example `C:\MATLAB\R2026a`. If you already have MATLAB R2026a installed, you can go to next step.

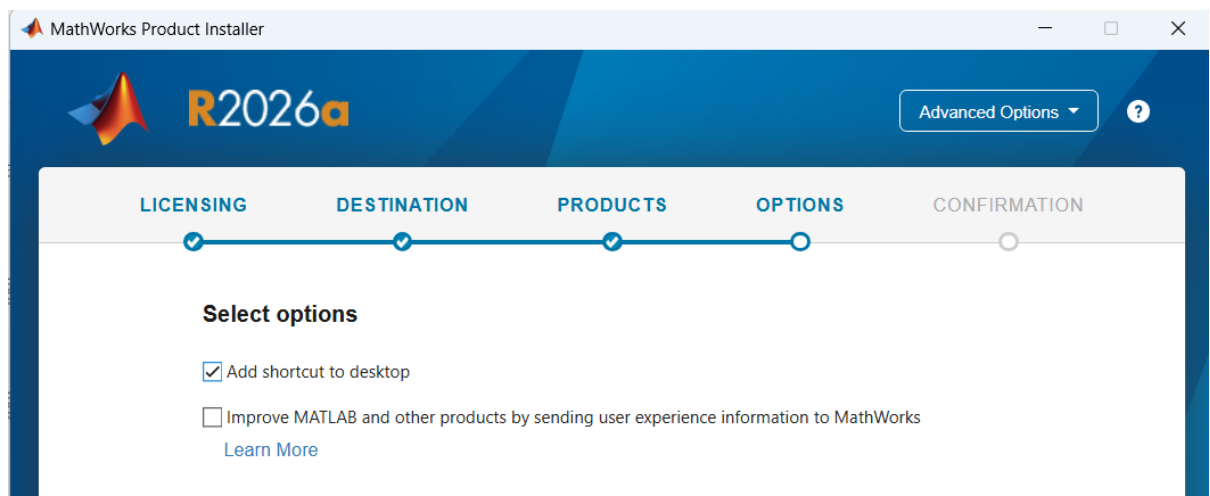


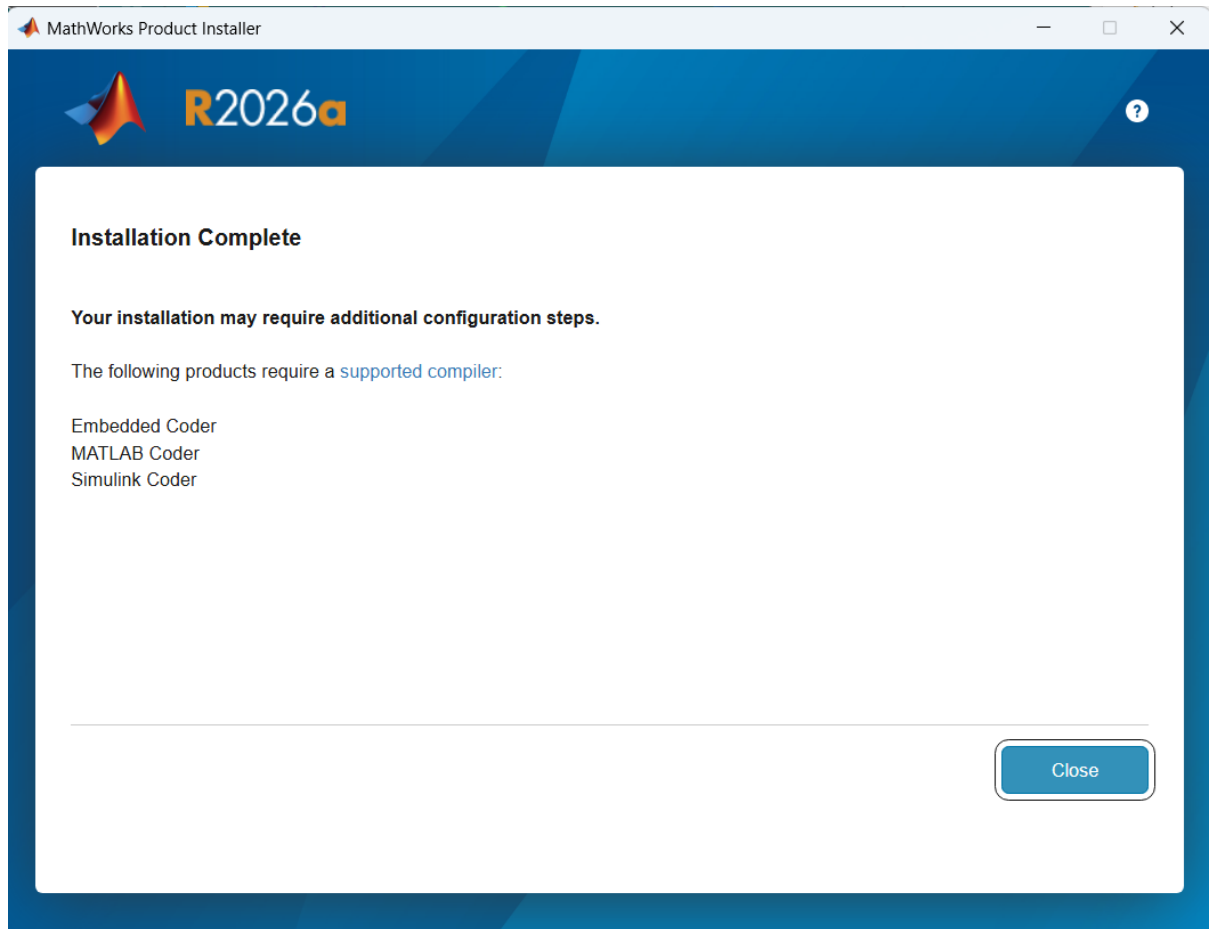


6. Select all the available toolboxes associated with the provided evaluation license, and proceed with the rest of the installation.



7. Select option **Add shortcut to desktop** and disable *Improve MATLAB and other products by sending user experience information to MathWorks*.





8. Once installed, please launch the **Activate MATLAB R2026a** recently installed, and follow the activation steps.

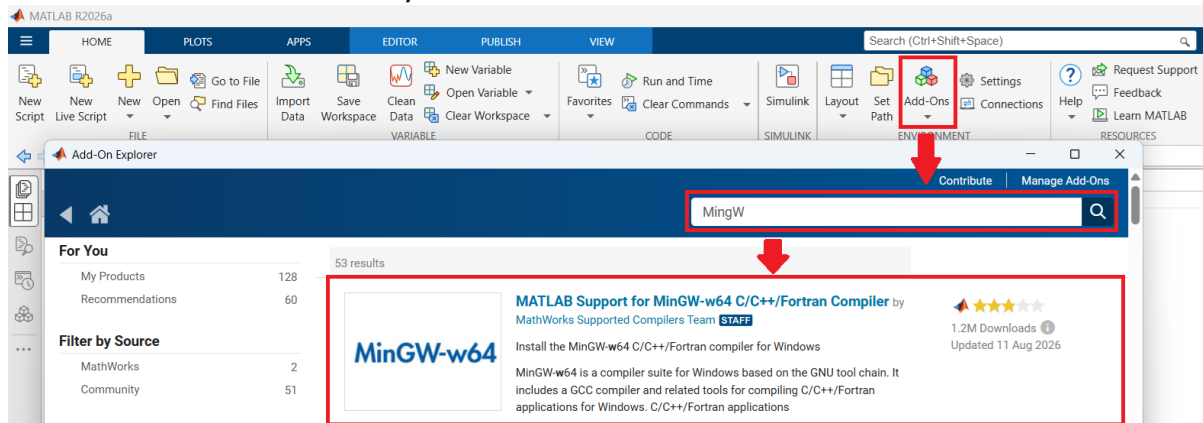
Note: when activating MATLAB R2026a, enter the following Activation Key if requested: 61237-04365-25622-63850-77394



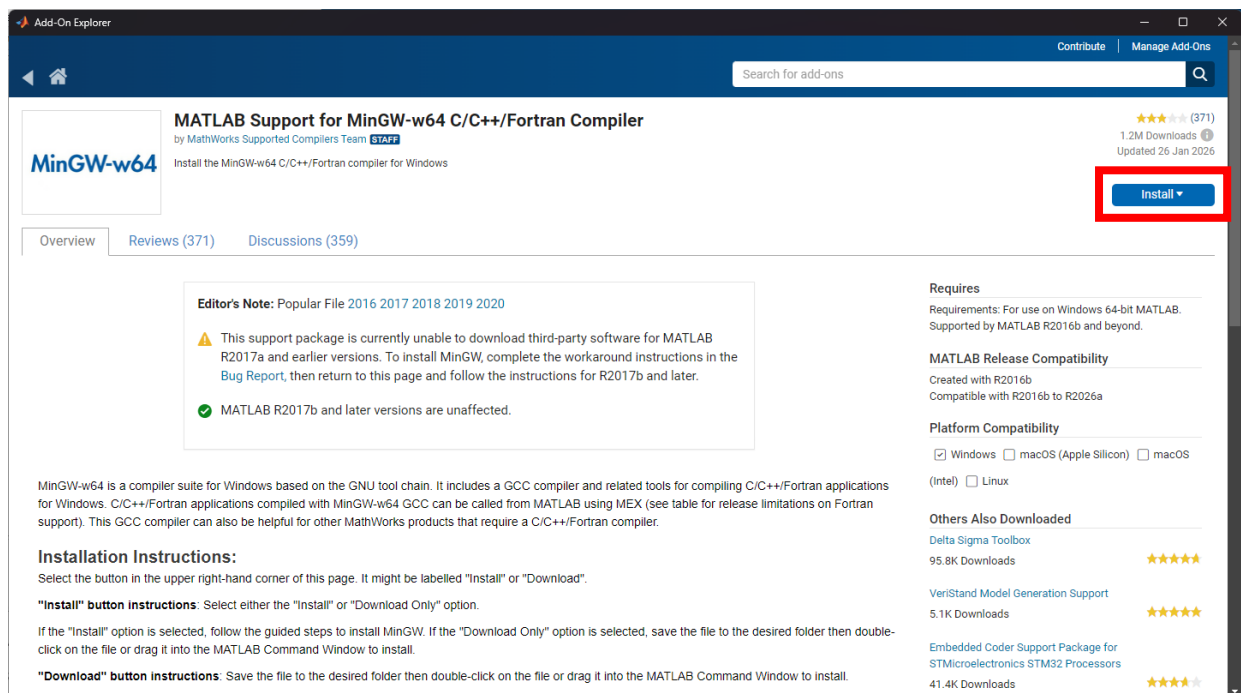
Install additional dependencies

MATLAB Support for MinGW-w64 C/C++/Fortran Compiler

Open the newly installed MATLAB R2026a. Go to MATLAB Add-Ons, and search for the **MinGW** key word.



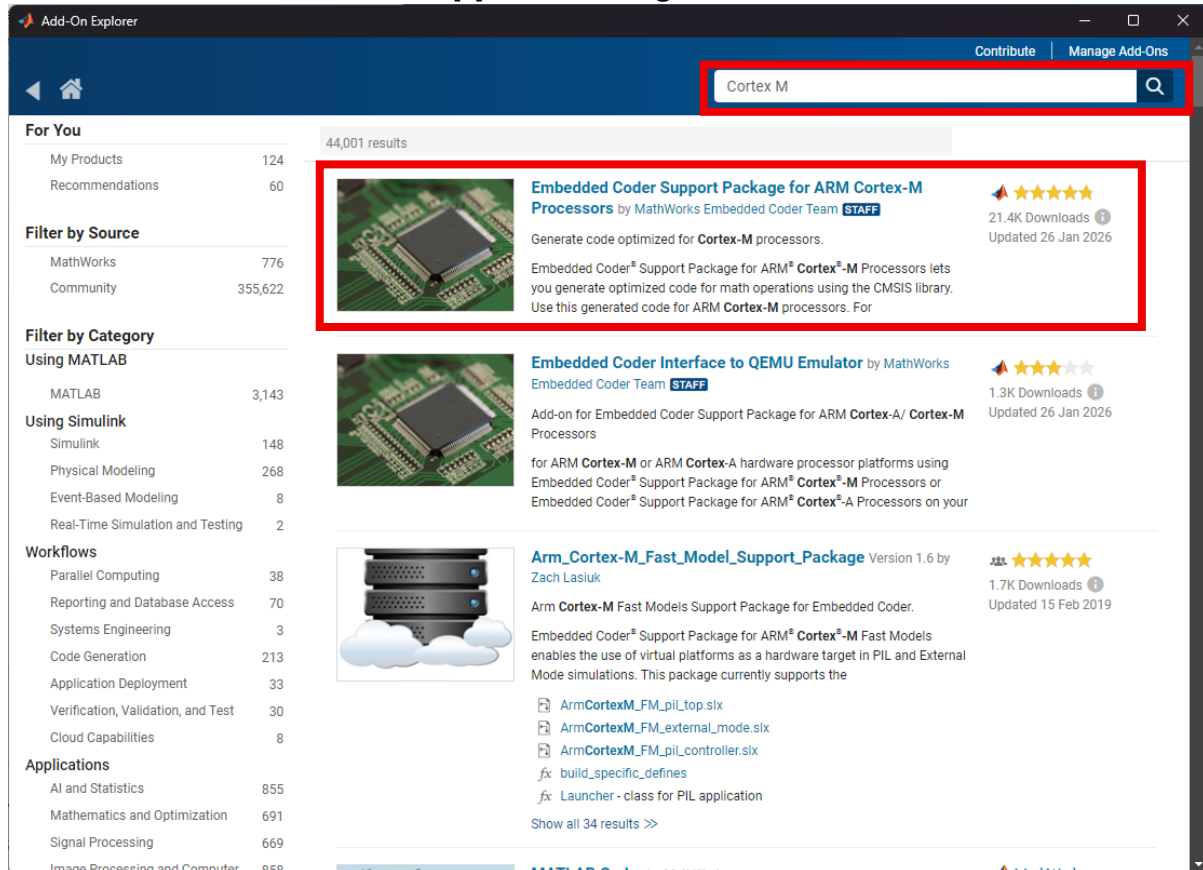
Open **MATLAB Support for MinGW-w64 C/C++/Fortran Compiler** package and press the **Install** button.



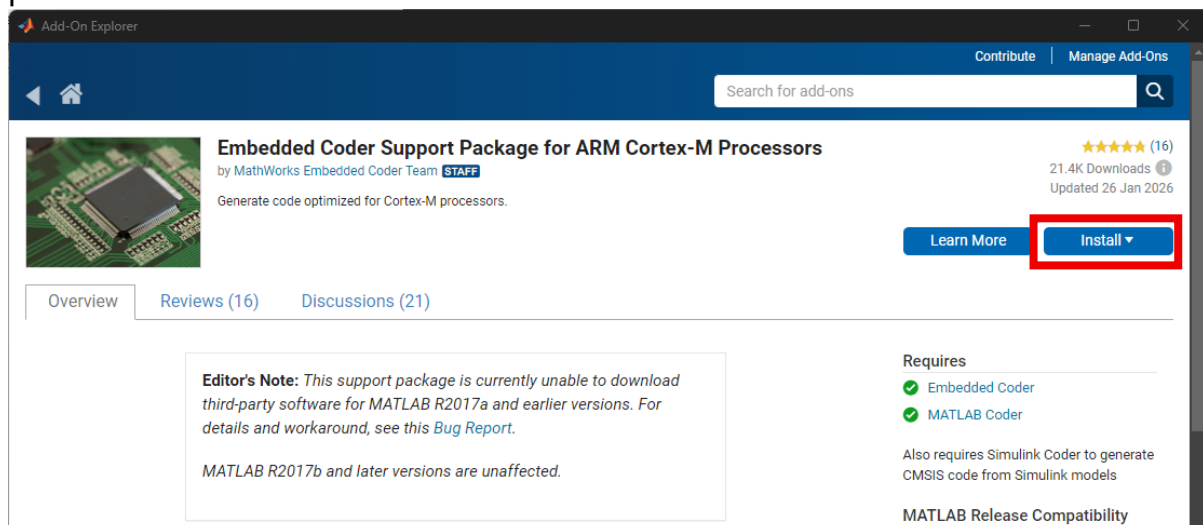


Embedded Coder Support Package for ARM Cortex-M Processors

Go to MATLAB Add-Ons, search for the **Cortex M** key word and scroll down for the **Embedded Coder Support Package for ARM Cortex-M**.



Open **Embedded Coder Support Package for ARM Cortex-M** package and press Install button.





2. Install Model-Based Design Toolbox for S32K3 1.8.0

Model-Based Design Toolbox for S32K3 enables rapid prototyping of complex algorithms on NXP S32K3 family of microcontrollers from MATLAB and Simulink.

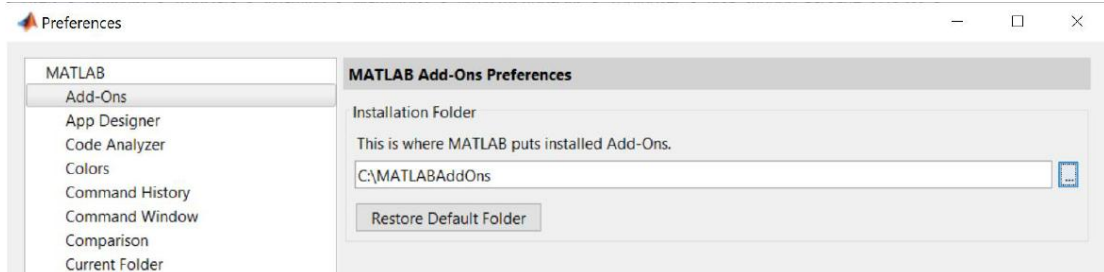
To install the Model-Based Design Toolbox from S32K3, please follow the steps described in the **SW32K3_MBDT_1.8.0_D2512_UserManual.pdf** document, chapter **1.2 Installation Steps (1.2.1 to 1.2.3)**.

As described in the attached document, MBDT for S32K3 is free of charge, no license is required.

Quick Notes and Suggestions

1. It is recommended to install the MATLAB and NXP Toolbox into a location that does not contains special characters, empty spaces, or mapped drives. Also, consider choosing a short path like *C:/MATLABAddOns*, like in the screenshot below.

The default location can be changed before installation by changing the Add-Ons path from MATLAB Preferences.



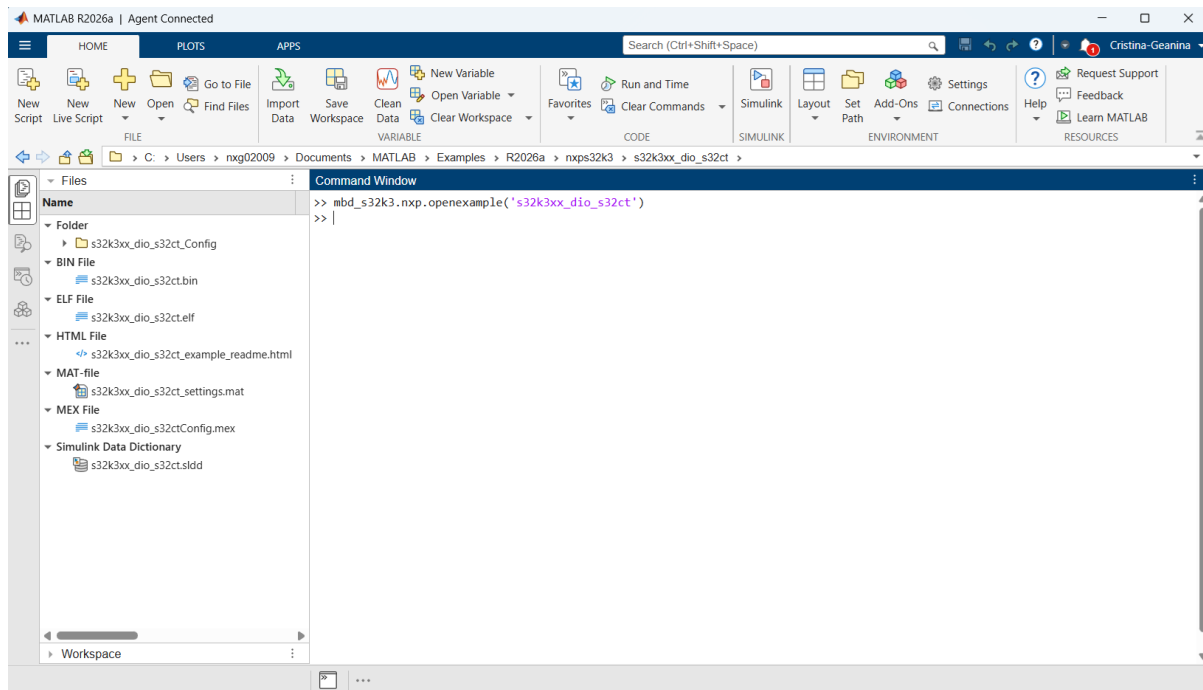
2. Installing EB tresos (chapter 1.2.4) is not required, since for the Hands-On we will use the S32 Configuration Tools as the hardware configuration tool. The S32 Configuration Tools is delivered within Model-Based Design Toolbox package. More details during the bootcamp.

Testing MBDT Installation

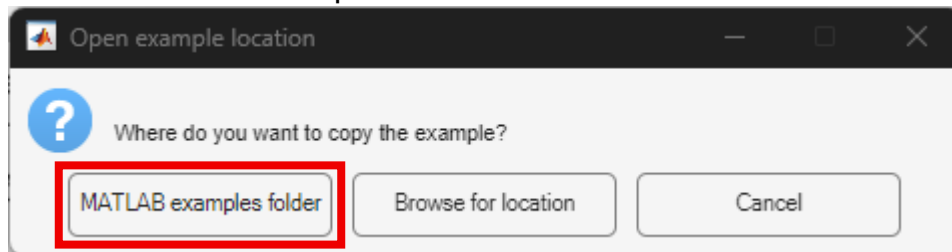
Once the MBDT for S32K3 is installed and activated, generate code for DIO model targeting the S32K344-MINI EVB.

1. In MATLAB Command Window, run the following command

```
>> mbd_s32k3.nxp.openexample('s32k3xx_dio_s32ct')
```

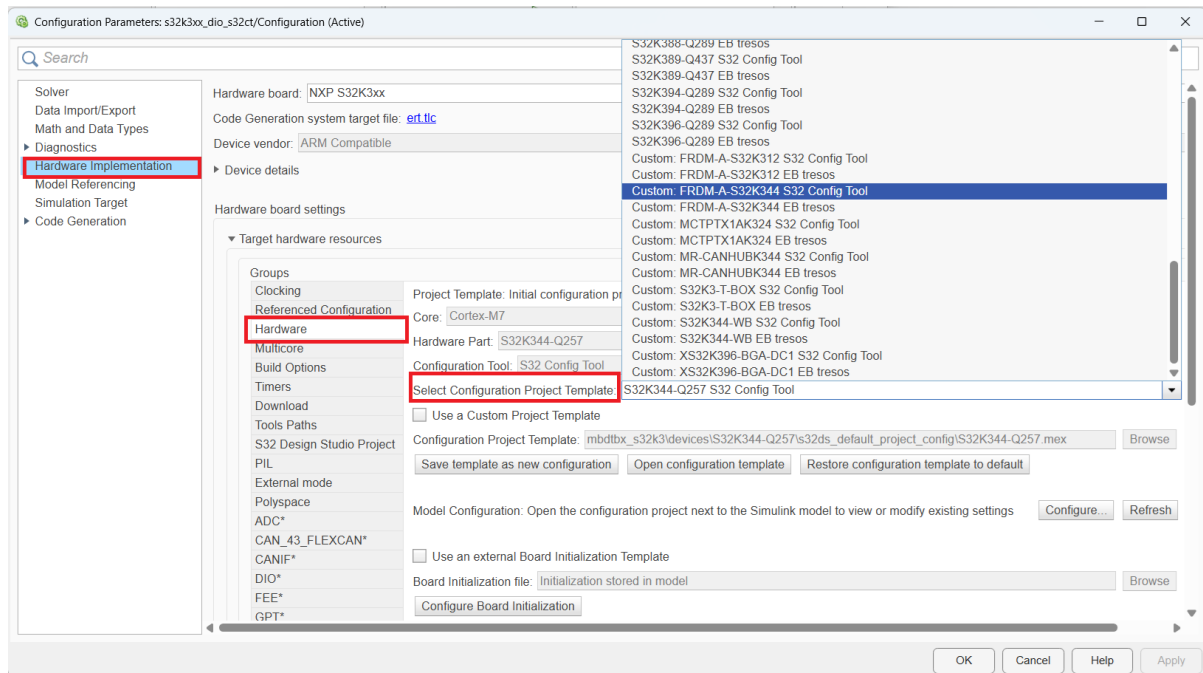


2. Select MATLAB examples folder.

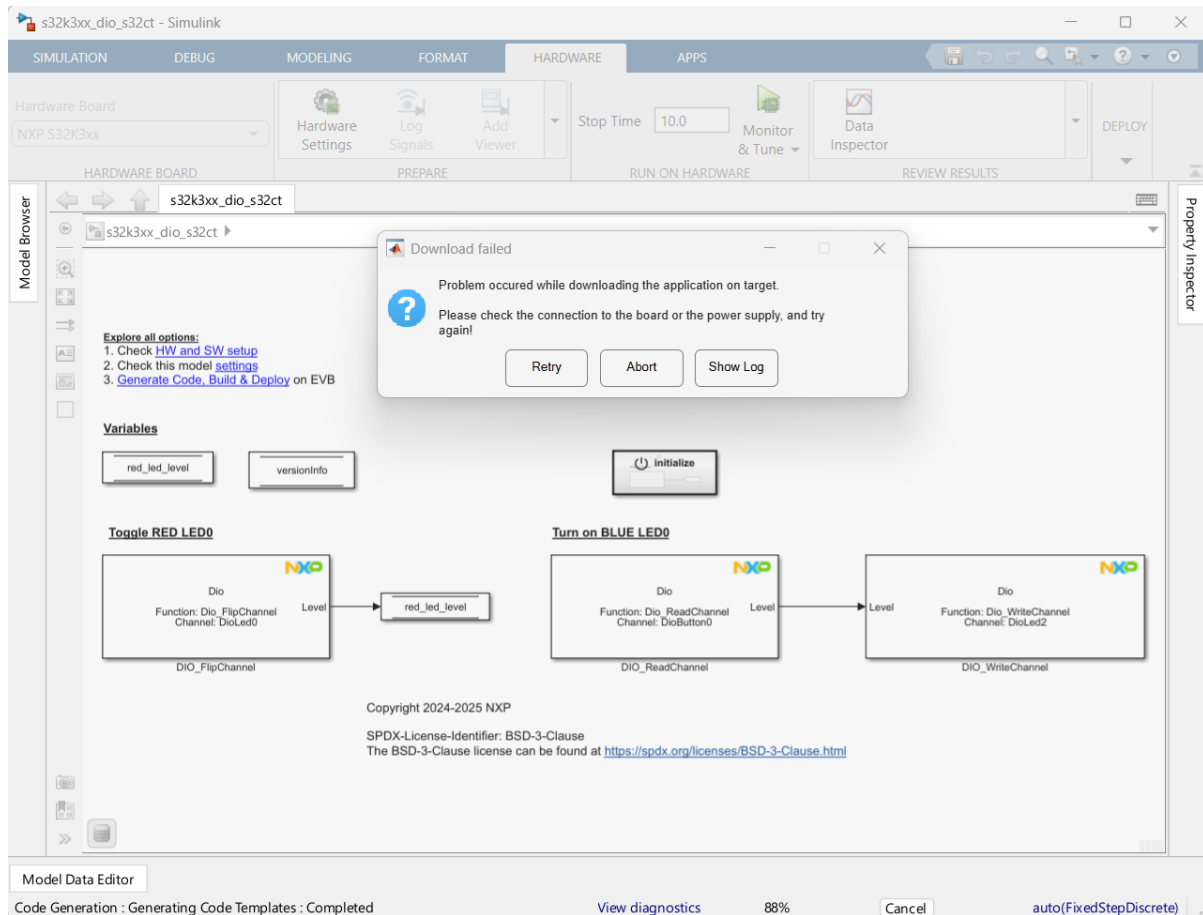


3. Open the Model Settings.

Under *Hardware Implementation* -> *Hardware* -> *Select Configuration Project Template* select **Custom: FRDM-A-S32K344 S32 Config Tool**. Then click **Apply** and **OK**.



4. Generate code for this model (Ctrl+B). If the code generation process is successful, a Download failed message appears, this means that the code has been generated successfully, but toolbox was unable to deploy the generated binary file on the board.

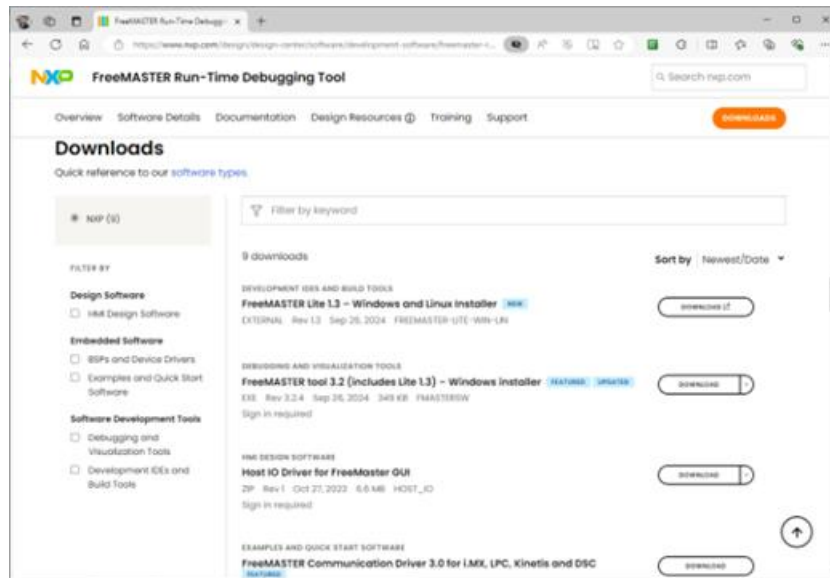


3. Install FreeMASTER

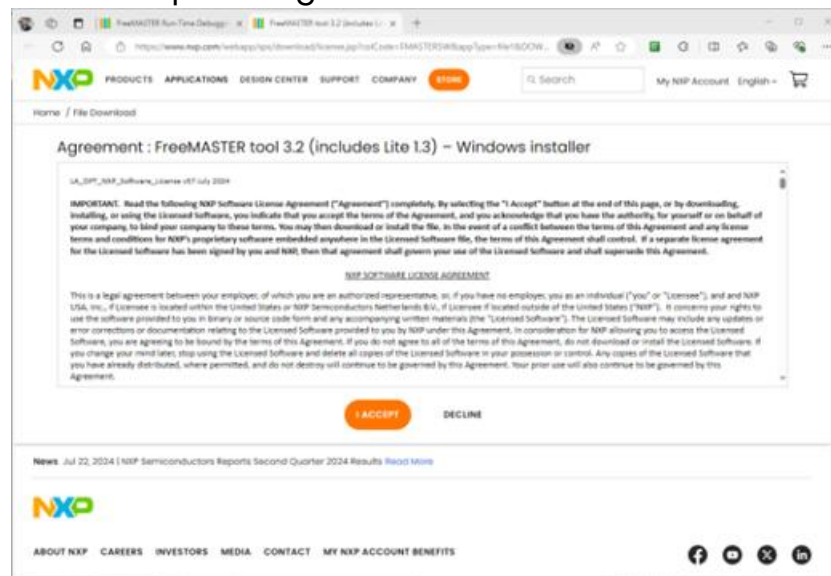
FreeMASTER is a user-friendly real-time debug monitor and data visualization tool that enables runtime configuration and tuning of embedded software applications.

To install FreeMASTER, please follow the steps below:

1. Go to FreeMASTER Run-Time Debugging Tool page accessing the following link: <https://www.nxp.com/design/design-center/software/development-software/freemaster-run-time-debugging-tool:FREEMASTER>
2. Go to Downloads Section, select the **FreeMASTER tool 3.2 (includes Lite 1.3) – Windows installer** and press the **Download** button. Sign in is Required.

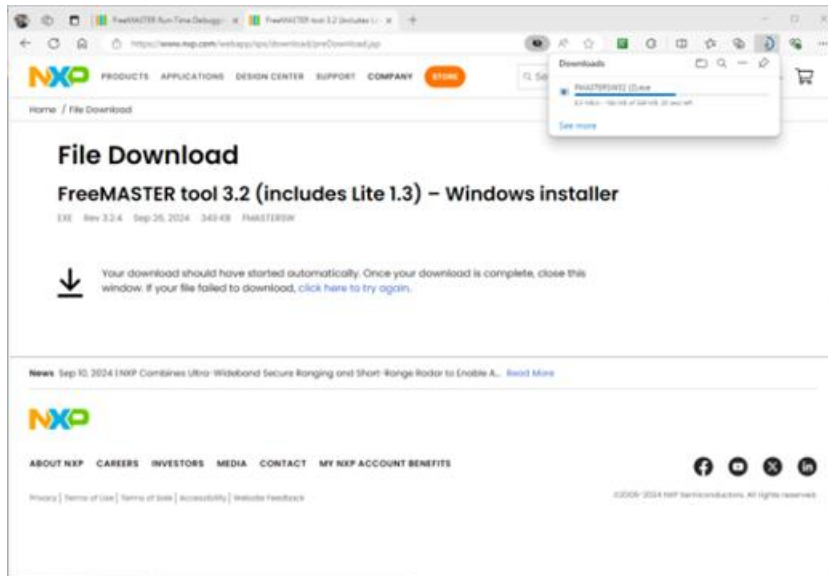


3. Review and accept the Agreement for the FreeMASTER tool.





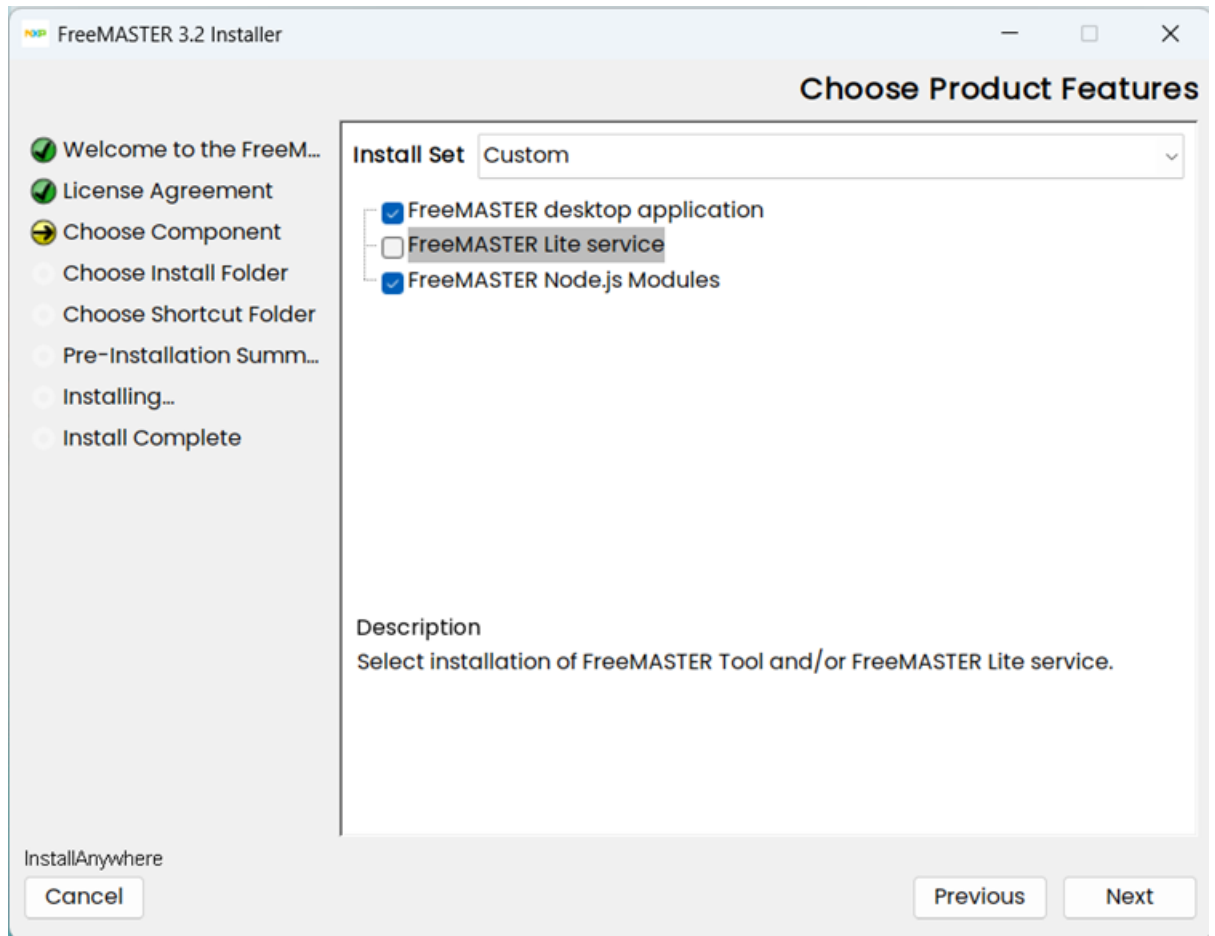
4. Download the **FMASTERSW32.exe** executable.



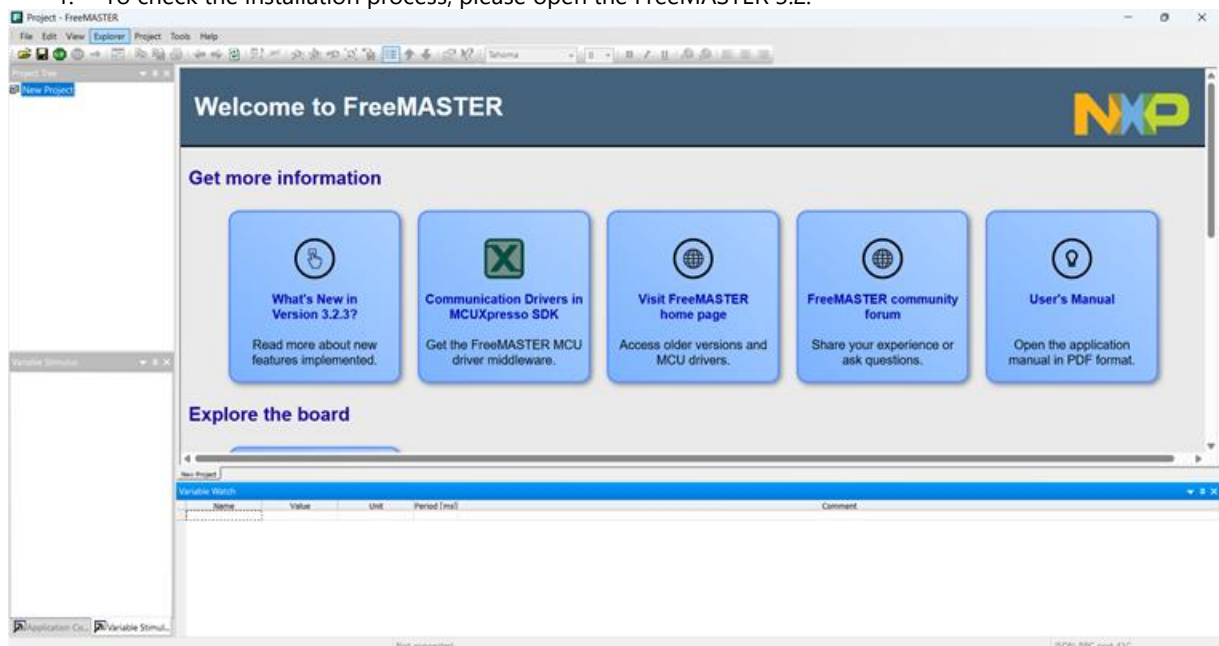
5. Run the **FMASTERSW32.exe** executable. Follow the Installation steps, by reading and agreeing the License.



6. At this step, unselect the installation of the FreeMASTER lite.



1. To check the installation process, please open the FreeMASTER 3.2.





4. Install PEmicro Drivers

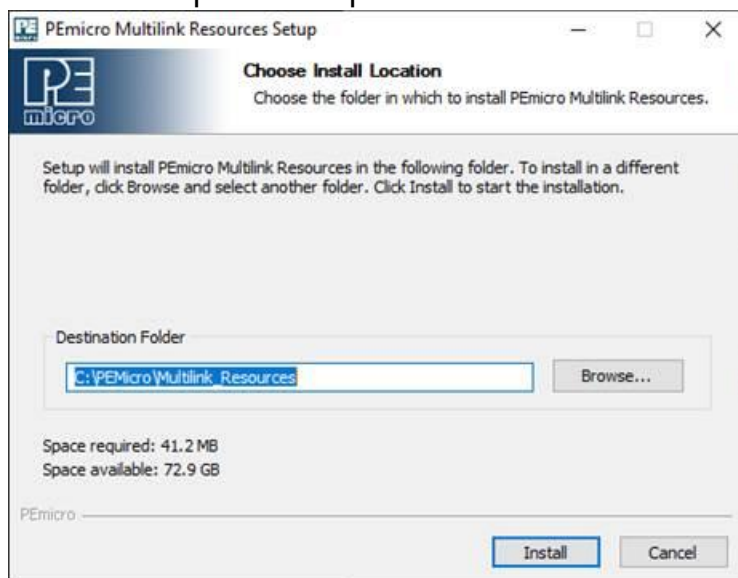
PEmicro drivers are required to be installed for the further use of the OpenSDA on board debugger/USB to serial converter.

To install the PEmicro Drivers, please follow the steps below:

1. Go to the following link and download the USB Multilink Resources Install.

https://www.pemicro.com/downloads/download_file.cfm?download_id=346

2. Run the **multilink_resources_install.exe**, keep the default installation path and press the **Install** button.



Now the pre-requisites are complete.